

The Boundaries of Inclusive Growth: Factor-Biased Technical Change Suitability, Human Capital, and Institutional Quality*

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Abstract

Does the alignment between technical change and factor endowments reduce income inequality only when economies can absorb and diffuse the gains? Using a balanced panel of 64 economies from 1980 to 2019, we show that capital-biased technical change suitability lowers the log market Gini by 0.82 percent per standard deviation. Human capital is the main amplifier: a one-standard-deviation increase in pre-sample schooling strengthens the equalizing effect by about 2.1 percentage points, primarily through global value chain upgrading and better employment composition. Institutional quality lowers inequality directly but does not linearly amplify the suitability channel; instead, weak institutions become binding only when human capital is also low. Two-stage least squares estimates using shift-share and spatial instruments confirm that these patterns reflect causal relationships rather than spurious correlations. These results imply that

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technology-factor alignment promotes inclusive growth only when paired with sustained investment in people and governance capacity.

JEL: O33, D31, J24, P16, F66

Keywords: biased technical change, income inequality, human capital, institutional quality, global value chains, factor endowments

1 Introduction

A central puzzle in development economics concerns why technological progress produces markedly different distributional outcomes across countries. Capital-intensive technologies imported by capital-scarce economies generate weaker employment and wage gains than identical technologies operating in capital-abundant settings [Acemoglu and Zilibotti, 2001, Caselli and Coleman, 2006]. Yet even after accounting for factor endowment mismatches, the equalizing effects of technically appropriate innovation remain heterogeneous. Some economies successfully translate factor-aligned technical change into broad-based income gains; others see productivity improvements appropriated by narrow elites. Understanding what determines which outcome prevails is central to designing technology policy that promotes inclusive growth.

This paper argues that two boundary conditions mediate whether capital-biased technical change suitability reduces income inequality: *human capital* and *institutional quality*. When the labor force is insufficiently educated, capital-complementary technologies raise productivity without expanding the set of tasks workers can perform alongside machines. The distributional benefits of technical change then accrue primarily to capital owners and a thin stratum of high-skill workers, leaving the broad middle of the wage distribution unaffected or worse off [Krusell et al., 2000, Moll et al., 2022]. When institutions are weak, efficiency gains are captured by politically connected insiders through rent-seeking rather than diffused via competitive labor markets and public goods provision [Acemoglu et al., 2005, Hall and Jones, 1999]. Both conditions choke off the channels through which factor-aligned technical change would otherwise compress wage dispersion.

We test this mechanism using variation in *capital-biased technical change suitability*—the degree to which an economy’s direction of technical change aligns with its capital-labor ratio—across 64 economies from 1980 to 2019. Our suitability measure follows Li et al. [2026] and Li et al. [2024], who construct suitability as the log ratio of the marginal products of capital and labor, a sufficient statistic for factor-direction alignment under standard constant-returns

production assumptions. Higher suitability means that the prevailing direction of capital-biased technical progress is more consistent with the economy’s relative factor abundance, reducing the implicit wedge between technology and endowments identified by [Acemoglu and Zilibotti \[2001\]](#).

Our empirical strategy exploits two-way fixed effects (TWFE) with Driscoll–Kraay standard errors [[Driscoll and Kraay, 1998](#)] that are robust to cross-sectional dependence and serial correlation, which are pervasive in cross-country panel data. We augment the TWFE baseline with interaction terms between suitability and (i) institutional quality, measured by the Fraser Institute’s Economic Freedom Index [[Gwartney et al., 2020](#)], and (ii) human capital, proxied by the log of mean schooling in 1970 [[Barro and Lee, 1996](#)]. The pre-sample human capital proxy reduces concerns about reverse causation from contemporary inequality to schooling investment. To address the potential endogeneity of suitability itself, we instrument using a shift-share (Bartik) instrument, the log mean suitability of other economies, and pre-sample capital-stock proxies, following the strategy in [Li et al. \[2026\]](#); for the moderation analysis we interact each instrument with the (exogenous) boundary variables.

The paper yields four main results. First, technical change suitability significantly reduces income inequality in the baseline specification. A one-standard-deviation increase in suitability (0.48 log points) is associated with a 0.82 percent decline in the log Gini coefficient, with the coefficient -0.017 significant at the 5 percent level. This confirms the aggregate finding in [Li et al. \[2026\]](#) and establishes it in the moderation framework of this paper.

Second, human capital is quantitatively the most important amplifier of the suitability–inequality relationship. The interaction between suitability and 1970 human capital carries a coefficient of -0.038 (significant at 1 percent), implying that moving from the median to one standard deviation above median human capital raises the equalizing effect of a one-unit increase in suitability by 2.1 percentage points. A high-human-capital dummy interaction confirms this nonlinearity.

Third, the linear interaction between suitability and institutional quality is negative but

statistically insignificant (-0.004 , $p > 0.10$). Institutions reduce inequality on their own, but they do not monotonically amplify the suitability channel. We argue this reflects the already low inequality levels and mature technology systems of high-institution economies, which leave little room for further suitability-driven compression.

Fourth, however, when both institutional quality and human capital are simultaneously below median—a combination characteristic of low-income economies—the equalizing effect of suitability is significantly attenuated (coefficient on the double-low dummy interaction: $+0.010$, significant at 10 percent). This threshold pattern explains a well-documented anomaly: in low-income economies, capital-biased technical change often *widens* income dispersion even when nominally appropriate for the factor endowment [Jaumotte et al., 2013].

Mechanism analysis clarifies *how* these boundary conditions operate. Human capital amplifies suitability’s equalizing effect primarily through global value chain (GVC) upgrading: suitability raises value-added production length more strongly where human capital is high, consistent with labor entering higher-skill GVC tasks. At the same time, high-human-capital economies show *less* expansion of vulnerable (informal, low-quality) employment in response to suitability improvements, suggesting that the equalizing mechanism shifts from quantity to quality of employment. Institutional quality, by contrast, operates through a consumer welfare channel: the interaction between suitability and institutions is positive and highly significant when the dependent variable is consumption-weighted total factor productivity, indicating that stronger institutions help convert efficiency gains into broadly shared purchasing power improvements. Together, these results reveal that human capital and institutional quality are *qualitatively different* boundary conditions: human capital is an absorption and amplification condition, while institutional quality is primarily a diffusion and floor condition.

Contributions. This paper makes three contributions to the literature. First, we extend the biased technical change literature—which has focused on the aggregate direction of innovation [Acemoglu, 2002, Acemoglu and Restrepo, 2018, 2020, 2022]—by introducing a

boundary-condition framework that explains cross-country heterogeneity in the distributional consequences of factor-aligned technical change. Second, we complement studies of technology and inequality [Jaumotte et al., 2013, Krusell et al., 2000, Moll et al., 2022, Karabarbounis and Neiman, 2014] by showing that the same variable—suitability—reduces inequality in human-capital-rich and institutionally strong economies but fails to do so when both conditions are absent simultaneously. Third, we provide cross-country evidence that GVC integration is a key mechanism linking human capital to the distributional benefits of technical change, adding to a growing literature on trade, technology, and inequality [Hornbeck and Moretti, 2024, Rodríguez and Rodrik, 2000, Timmer et al., 2015].

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework and states four testable hypotheses. Section 3 describes the data, variables, and empirical strategy. Section 4 presents the main regression results. Section 5 reports the mechanism analysis. Section 6 presents robustness and identification tests. Section 7 concludes with policy implications.

2 Theoretical Framework and Hypotheses

2.1 Biased Technical Change Suitability and Inequality

Acemoglu and Zilibotti [2001] show that technologies developed for high capital-labor-ratio economies raise productivity less when transplanted to labor-abundant developing countries. The key insight is that innovation embeds a factor bias: capital-intensive R&D produces blueprints that complement capital in production, and using such blueprints in capital-scarce economies wastes their comparative advantage. Li et al. [2026] and Li et al. [2024] formalize *suitability* as the realized alignment between factor bias and factor endowments: it equals the log marginal product ratio $\ln(\partial Y/\partial K) - \ln(\partial Y/\partial L)$, which captures how closely an economy’s technical trajectory matches its actual capital-labor ratio.

When suitability is high, capital-biased technical change amplifies total factor productivity

through three channels. *Channel 1 (GVC upgrading)*: higher suitability enables firms to enter more capital-intensive, higher-value-added segments of global production networks, redistributing rents from foreign lead firms toward domestic labor. *Channel 2 (employment quality)*: factor alignment shifts production toward tasks where labor and capital are complements rather than substitutes [Autor et al., 2003], raising the relative demand for middle-skill workers and attenuating wage polarization. *Channel 3 (welfare diffusion)*: efficiency gains lower consumption costs and improve access to public goods, raising real incomes at the lower end of the distribution [Atkinson, 2015].

Hypothesis 1. *Capital-biased technical change suitability reduces income inequality.*

2.2 Human Capital as the Absorption Boundary

The three channels above are all contingent on workers being able to enter new tasks alongside capital. When human capital is high, workers possess the cognitive and technical skills to operate, adapt, and improve new equipment. Firms adopting capital-intensive technologies then need educated workers to manage and augment machines, raising skilled-labor demand and compressing the skill premium via *capital-skill complementarity* [Krusell et al., 2000]. The GVC channel is especially human-capital-intensive: moving into higher-value-added production stages requires organizational capability, process engineering, and managerial depth that low-education labor forces lack [Goldin and Katz, 2009].

When human capital is low, the reverse applies. Workers cannot enter technologically sophisticated tasks, so capital-biased technical change raises capital returns without commensurate gains for labor. Tasks that could in principle be upgraded remain off-limits to the bulk of the workforce, and the employment margin shifts toward informal, low-quality jobs. The GVC channel stalls because firms cannot climb the value ladder without educated workers. Consequently, suitability’s equalizing power depends critically on the economy’s human capital stock.

Hypothesis 2. *Higher human capital amplifies the income-equalizing effect of capital-biased technical change suitability.*

2.3 Institutional Quality as the Diffusion Boundary

Even when suitability raises economy-wide productivity and human capital enables workers to enter new tasks, distributional benefits may not materialize if institutions are weak. Strong institutions—secure property rights, competitive markets, transparent governance, and robust social insurance—limit rent-seeking, protect workers’ bargaining power, and channel public revenues from growth into health, education, and welfare [Acemoglu et al., 2005, Hall and Jones, 1999]. In such environments, the efficiency gains from factor-aligned technical change flow broadly: firms compete away rents, governments can tax and redistribute, and workers have legal recourse against monopsonistic wage suppression.

Conversely, when institutions are weak, elites and politically connected firms can appropriate the surplus generated by suitability improvements [?]. Tax systems are too thin to redistribute; public goods provision stagnates; labor regulations go unenforced. The consumer welfare channel is especially sensitive: turning aggregate efficiency gains into lower consumer prices and better public services requires fiscal capacity and rule of law.

Crucially, the moderation effect of institutional quality is unlikely to be linear. High-institution economies already exhibit compressed inequality and mature technological systems, so the marginal distributional gain from further suitability improvements is small. The critical role of institutions may be as a *floor condition*: preventing the worst distributional outcomes when suitability rises in economies where human capital is also insufficient.

Hypothesis 3. *Institutional quality acts as a threshold rather than a linear amplifier: the equalizing effect of suitability is significantly attenuated only when both institutional quality and human capital are simultaneously below the median.*

2.4 Channels of Boundary Conditions

The two boundary conditions engage distinct mechanisms. Human capital expands the set of workers who can perform the high-skill tasks associated with GVC upgrading, so the primary human-capital-mediated channel should be value-added GVC integration. Institutional quality, by contrast, determines whether efficiency gains are converted into public goods and lower consumer prices, so the primary institution-mediated channel should be consumer welfare improvement.

Hypothesis 4. *Human capital amplifies suitability’s equalizing effect primarily through global value chain upgrading, while institutional quality operates primarily through consumer welfare diffusion.*

3 Data, Variables, and Empirical Strategy

3.1 Sample and Data Sources

Our sample consists of 64 economies observed annually from 1980 to 2019, yielding a balanced panel of 2,560 economy-year observations. The dataset is constructed by [Li et al. \[2026\]](#) and integrates five primary sources: (i) the Standardized World Income Inequality Database (SWIID) [\[Solt, 2016\]](#) for income inequality; (ii) Penn World Table version 10.0 [\[Feenstra et al., 2015\]](#) for output and factor inputs; (iii) the Fraser Institute’s Economic Freedom Index [\[Gwartney et al., 2020\]](#) for institutional quality; (iv) historical schooling data from Clio Infra for the 1970 human capital proxy; and (v) the UIBE Global Value Chain Database for GVC indicators.

We extend the original empirical program of [Li et al. \[2026\]](#) by adding moderation specifications without altering the outcome variable, core explanatory variable, controls, or fixed-effect structure. This design maximizes comparability with the baseline estimates.

3.2 Variables

Income inequality ($\ln \text{Gini}_{it}$). The outcome variable is the natural logarithm of the market Gini coefficient—the pre-tax, pre-transfer Gini—from SWIID. Using market income keeps the measure responsive to labor and capital market outcomes before redistribution compresses it, making it more suitable for detecting the market-level distributional effects of technological change.

Technical change suitability ($\ln \text{Suit}_{it}$). The core explanatory variable is the log ratio of the marginal product of capital to the marginal product of labor, constructed by [Li et al. \[2024\]](#). Formally:

$$\ln \text{Suit}_{it} = \ln(\text{MPK}_{it}) - \ln(\text{MPL}_{it}), \quad (1)$$

where marginal products are derived from a CES production function estimated at the economy level. Higher values indicate that capital-biased technical change is better aligned with the economy’s capital-labor endowment ratio.

Institutional quality (EFI_{it}). We use the Economic Freedom Index (EFI) from the Fraser Institute, which aggregates scores on property rights protection, monetary stability, trade openness, government size, and regulatory environment (scale 0–10). Higher EFI values indicate stronger market-supporting institutions.

Human capital ($\ln \text{HC}_i$). We proxy human capital using the log of mean years of schooling in 1970, from Clio Infra [\[Li et al., 2026\]](#). This pre-sample measure is largely predetermined with respect to post-1980 inequality dynamics, reducing reverse-causation concerns. The cost is that it cannot capture within-sample human capital changes; we address this limitation in Section 6 by using the dynamic PWT human capital index as an alternative.

Control variables. We follow [Li et al. \[2026\]](#) in controlling for trade openness (tra, exports plus imports as a share of GDP), institutional quality (as above), the old-age dependency

ratio (old), and log population (ln pop). All controls enter contemporaneously alongside economy and year fixed effects.

3.3 Descriptive Statistics

Table 1 reports descriptive statistics for all variables. The log Gini has a mean of 3.69 and a standard deviation of 0.28, reflecting substantial cross-country and over-time variation. Suitability averages 0.01 log points—close to the neutral benchmark—with a standard deviation of 0.48, spanning from -2.33 for highly labor-intensive economies to $+1.88$ for capital-intensive ones. Institutional quality averages 6.68 on a 0–10 scale, with a standard deviation of 1.26. The 1970 schooling proxy has a mean of 1.78 log years and standard deviation of 0.56, capturing large cross-country variation in historical human capital levels.

Table 1: Descriptive Statistics

Variable	Description	Obs.	Mean	Std. Dev.	Min	Max
ln Gini	Log income inequality (market Gini)	2,560	3.6884	0.2847	2.8745	4.3449
ln Suit	Log biased TC suitability	2,560	0.0103	0.4799	-2.3347	1.8755
EFI	Institutional quality (Fraser EFI)	2,472	6.6795	1.2648	3.1100	9.1100
ln HC ₇₀	Human capital (log schooling, 1970)	2,280	1.7800	0.5634	0.0951	2.5006
old	Old-age dependency ratio	2,560	0.1416	0.0813	0.0273	0.4852
tra	Trade openness (% GDP)	2,526	81.44	63.06	9.14	442.62
ln pop	Log population (millions)	2,560	2.5210	1.7846	-1.4773	7.2681

Notes: Sample is a balanced panel of 64 economies, 1980–2019. TC = technical change. EFI = Economic Freedom Index (Fraser Institute, scale 0–10). Obs. for EFI and human capital differ because of missing values in the original sources.

3.4 Empirical Strategy

Baseline specification. We estimate:

$$\ln \text{Gini}_{it} = \alpha + \beta \ln \text{Suit}_{it} + \boldsymbol{\gamma}' \mathbf{X}_{it} + \lambda_t + \theta_i + \varepsilon_{it}, \quad (2)$$

where λ_t are year fixed effects, θ_i are economy fixed effects, and $\mathbf{X}_{it}$ is the vector of controls. Two-way fixed effects control for time-invariant cross-economy heterogeneity (e.g. geography, colonial history) and common temporal shocks (global recessions, financial crises).

Moderation specifications. To test Hypotheses 2–4, we add an interaction term:

$$\ln \text{Gini}_{it} = \alpha + \beta_1 \ln \text{Suit}_{it} + \beta_2 (\ln \text{Suit}_{it} \times \text{Boundary}_{it}) + \boldsymbol{\gamma}' \mathbf{X}_{it} + \lambda_t + \theta_i + \varepsilon_{it}, \quad (3)$$

where Boundary is alternately: (a) the mean-centered EFI (institutional quality); (b) the mean-centered 1970 log schooling (human capital); (c) a dummy equal to one for economies above median country-mean human capital (“high-HC group”); and (d) a dummy equal to one for economies simultaneously below median in both institutional quality and human capital (“double-low”). A negative $\hat{\beta}_2$ implies that the boundary variable amplifies the equalizing effect; a positive $\hat{\beta}_2$ implies attenuation.

Identification and instruments. Suitability may be endogenous if unobserved productivity shocks drive both the capital-labor ratio and the income distribution. Following [Li et al. \[2026\]](#), we address this concern using four instruments for $\ln \text{Suit}_{it}$: (Z1) the third and fourth lags of the log capital stock per worker ($\ln k_{it-3}$, $\ln k_{it-4}$), which predetermine the factor-intensity trajectory without being affected by contemporaneous distributional shocks; (Z2) a Bartik shift-share instrument that interacts economy-level sectoral employment shares in a base year with global sectoral capital-deepening trends, isolating variation in suitability driven by global rather than domestic forces; (Z3) the log mean suitability of all

other economies in the sample ($\ln \overline{\text{Suit}}_{-i,t}$), which proxies for technology frontier effects while being excludable because a single economy’s inequality does not drive the world technology frontier. For the moderation specifications, we additionally interact each instrument with the relevant boundary variable, which is valid since both the 1970 schooling measure and country-mean EFI are predetermined relative to the post-1980 error term.

Inference. Standard errors follow [Driscoll and Kraay \[1998\]](#), which are consistent under arbitrary cross-sectional dependence and $MA(q)$ serial correlation; we set the Bartlett kernel bandwidth to three periods. This procedure is more conservative than cluster-robust standard errors and is better suited to panels with moderate T relative to N [[Driscoll and Kraay, 1998](#)]. All continuous moderators are demeaned before constructing interaction terms so that the coefficient on $\ln \text{Suit}_{it}$ retains its interpretation as the average partial effect at the sample mean of the moderator.

4 Main Results

4.1 Baseline Effect and Boundary Conditions

Table [2](#) reports results from equations (2) and (3).

Table 2: Capital-Biased Technical Change Suitability and Income Inequality: Boundary Conditions

	(1)	(2)	(3)	(4)	(5)
	Baseline	Institution moderator	Human capital moderator	High-HC group	Double-Low group
$\ln \text{Suit}$	-0.0171^{**} (0.0078)	-0.0202^{**} (0.0082)	-0.0351^{***} (0.0108)	-0.0139^* (0.0083)	-0.0270^{***} (0.0102)
$\ln \text{Suit} \times \text{EFI}$		-0.0036 (0.0051)			
$\ln \text{Suit} \times \ln \text{HC}_{70}$			-0.0381^{***} (0.0082)		
$\ln \text{Suit} \times \mathbf{1}[\text{High-HC}]$				-0.0244^{***} (0.0082)	
$\ln \text{Suit} \times \mathbf{1}[\text{Double-Low}]$					0.0100^* (0.0056)
Observations	2,438	2,438	2,158	2,158	2,158
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is $\ln \text{Gini}_{it}$ (log market Gini coefficient). All regressions include economy and year fixed effects. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. EFI and $\ln \text{HC}_{70}$ are demeaned before constructing interaction terms. $\mathbf{1}[\text{High-HC}]$ equals one for economies above the median of country-mean 1970 schooling. $\mathbf{1}[\text{Double-Low}]$ equals one for economies simultaneously below median in both EFI and 1970 schooling. Standard errors in parentheses are Driscoll–Kraay [Driscoll and Kraay, 1998] with Bartlett kernel bandwidth of 3, robust to cross-sectional dependence and serial correlation. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Column (1) estimates the baseline specification without interactions. The coefficient on $\ln \text{Suit}$ is -0.0171 and significant at the 5 percent level (standard error 0.0078). Given that the standard deviation of $\ln \text{Suit}$ is 0.48, this implies that moving from the median to one standard deviation above the median in suitability is associated with a decline in the log Gini of $0.48 \times 0.0171 \approx 0.82$ percent—a meaningful reduction in market income dispersion. Hypothesis 1 is supported.

Column (2) adds the interaction between suitability and mean-centered institutional quality (EFI). The main effect of suitability remains negative and significant (-0.0202 , $p < 0.05$), but the interaction coefficient is -0.0036 with a standard error of 0.0051, which is statistically indistinguishable from zero. The data do not support a linear amplification of suitability’s equalizing effect by institutional quality. As discussed in Section 2, this null result is consistent with diminishing returns in high-institution economies: EFI independently reduces inequality ($\hat{\gamma}_{EFI} < 0$ in all specifications), but the marginal contribution of institutions to the suitability channel is not detectable at conventional significance levels.

Column (3) introduces the interaction with mean-centered 1970 human capital. The interaction coefficient is -0.0381 , significant at the 1 percent level (standard error 0.0082). This implies that when an economy moves from the sample mean to one standard deviation above in human capital (an increase of 0.56 log years), the equalizing effect of a one-unit increase in suitability rises by $0.56 \times 0.038 \approx 2.1$ percentage points. The suitability main effect remains strongly negative (-0.0351 , $p < 0.01$), indicating robust equalizing effects even at mean human capital. Hypothesis 2 is supported.

Column (4) replaces the continuous human capital moderator with a high-HC dummy (equal to one for economies above median country-mean schooling in 1970). The interaction coefficient is -0.0244 ($p < 0.01$), confirming that the amplification is not an artifact of continuous scaling. High-human-capital economies experience an additional 2.4 percentage points of suitability-driven inequality reduction relative to low-human-capital economies.

Column (5) tests the double-low threshold in Hypothesis 3, replacing the moderator

with a dummy for economies simultaneously below median in both institutional quality and human capital. The interaction coefficient is +0.0100, significant at the 10 percent level. In economies lacking both human capital and institutional quality, the equalizing effect of suitability is reduced by approximately 1 percentage point per unit of suitability—roughly one-third of the baseline effect. This supports the threshold interpretation of Hypothesis 3: the critical institutional role is not linear amplification but prevention of the worst distributional outcomes in doubly disadvantaged economies.

4.2 Economic Magnitude and Interpretation

The results in Table 2 reveal an asymmetry between the two boundary conditions. Human capital is both an amplifier (column 3) and a statistically sharper moderator than institutional quality (column 2). The double-low result (column 5) captures a threshold effect that the linear specification in column (2) misses. Together, these findings paint a coherent picture: *human capital is the absorption condition*—without it, the productivity gains from factor alignment cannot propagate into labor markets broadly—while *institutional quality is the diffusion floor*—without it, gains concentrate rather than spread, but the constraint is binding only when human capital is also weak.

Figure 1 illustrates the conceptual framework: suitability raises economy-wide productivity, but the translation into distributional improvement requires passing through human-capital and institutional filters. Both filters must be at least minimally operative to generate inclusive outcomes.

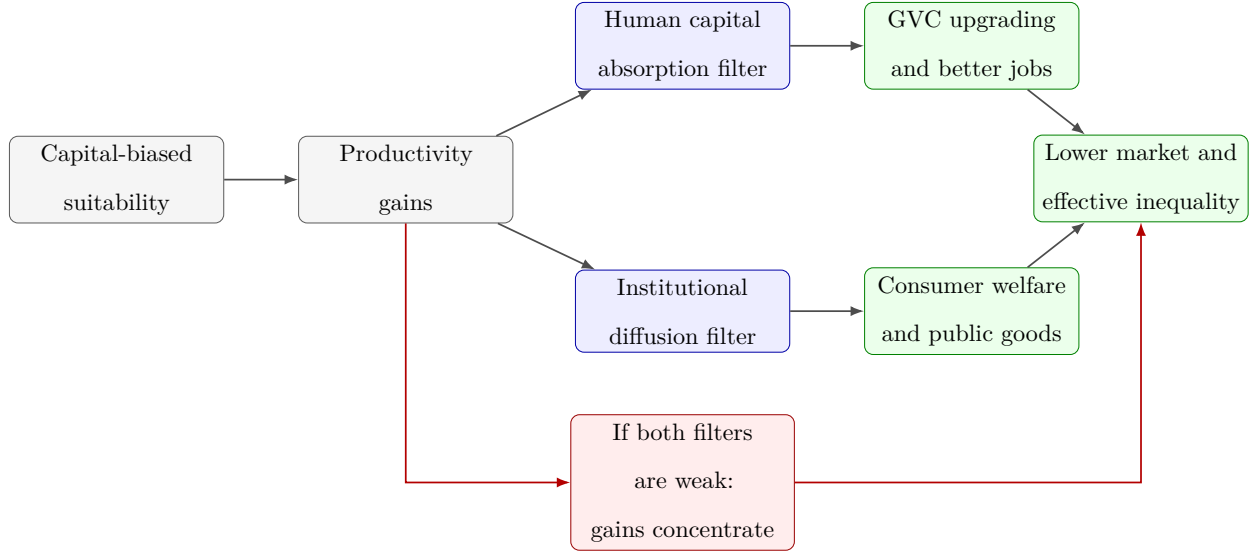


Figure 1: Conceptual Framework: Boundary Conditions on the Suitability–Inequality Channel

5 Mechanism Analysis

To test Hypothesis 4, we examine whether boundary conditions amplify suitability’s effect through the three channels identified in Section 2: GVC upgrading, employment quality, and consumer welfare. We estimate equation (3) with alternative outcome variables, reporting results in Table 3.

Table 3: Mechanism Analysis: Through Which Channels Do Boundary Conditions Operate?

	(1)	(2)	(3)	(4)	(5)
	GVC	Value-Added	Labor	Vulnerable	Consumer
	Position	Production	Particip.	Employment	Welfare
		Length	Rate	Rate	(CWTFP)
ln Suit	0.0839*** (0.0217)	0.0361* (0.0217)	0.4291*** (0.1478)	1.0287*** (0.3246)	0.0436*** (0.0074)
ln Suit $\times$ ln HC ₇₀	0.0151 (0.0281)	0.0539*** (0.0140)	-1.5574** (0.6397)	-4.0478*** (0.7131)	
ln Suit $\times$ EFI					0.0268*** (0.0055)
Observations	888	888	1,513	1,585	2,438
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variables are as listed in column headers. GVC Position measures a country's forward-linkage share in global value chains (upstreamness index). Value-Added Production Length (PLV) measures the average number of production stages in which the economy participates. Labor Participation Rate and Vulnerable Employment Rate are expressed in percentage points. Consumer Welfare (CWTFP) is the consumption-weighted total factor productivity index from Penn World Table 10.0 [Feenstra et al., 2015]. All regressions include economy and year fixed effects. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. Continuous moderators are demeaned. Standard errors in parentheses are Driscoll–Kraay [Driscoll and Kraay, 1998] with bandwidth 3. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.1 Human Capital and GVC Upgrading

Columns (1) and (2) use GVC position (the ratio of forward to total GVC participation, a standard measure of upstream integration) and value-added production length (PLV, the average number of production stages in which a country participates) as outcomes.

Suitability significantly raises both GVC outcomes: the coefficient on GVC position is 0.084 ($p < 0.01$) and on PLV is 0.036 ($p < 0.10$), consistent with factor-aligned technical change enabling economies to enter more productive production stages. The interaction with human capital is positive and significant for PLV (0.054, $p < 0.01$) but insignificant for GVC position (0.015, $p > 0.10$). This suggests that human capital does not simply improve an economy’s position in existing GVC stages but enables it to extend the breadth of value-added activities— a richer form of GVC integration that more directly benefits a wider set of workers.

5.2 Human Capital and Employment Quality

Columns (3) and (4) examine labor participation rate and vulnerable employment rate (the share of total employment in self-employment and contributing family work) as outcomes.

Suitability significantly raises both the participation rate (0.429, $p < 0.01$) and the vulnerable employment rate (1.029, $p < 0.01$). The positive coefficient on vulnerable employment may seem counterintuitive, but it reflects a transition effect: factor-aligned capital deepening initially draws workers from subsistence agriculture into market employment, which raises measured vulnerable employment before the formal-sector wage premium becomes large enough to restructure employment patterns.

The human capital interaction with both labor participation (-1.557 , $p < 0.05$) and vulnerable employment (-4.048 , $p < 0.01$) is strongly negative. This pattern is theoretically coherent: in high-human-capital economies, the employment margin of suitability improvement lies in *quality* (formal, higher-skill, higher-wage employment) rather than *quantity* (expanding any form of market participation). High-human-capital economies already have near-full

labor force participation, so suitability improvements redirect workers from vulnerable toward formal employment rather than expanding the extensive margin. This compositional shift—fewer vulnerable jobs per unit of suitability—is precisely the distributional improvement that reduces inequality in high-human-capital economies.

5.3 Institutional Quality and Consumer Welfare

Column (5) uses consumption-weighted total factor productivity (CWTFP) as a proxy for consumer welfare, following the approach in [Feenstra et al. \[2015\]](#). Suitability significantly raises CWTFP (0.044, $p < 0.01$). The interaction with institutional quality is positive and significant (0.027, $p < 0.01$). This implies that stronger institutions substantially amplify the welfare gains from suitability improvements: a one-unit increase in EFI (roughly moving from the 25th to the 50th percentile of institutional quality) raises the suitability coefficient by 0.027, approximately 60 percent of the baseline suitability effect.

This institutional amplification of the welfare channel explains why institutional quality does not appear as a significant linear moderator in the inequality regressions of Table 2. Institutional quality primarily converts efficiency gains into consumer price reductions and public goods improvements—outcomes that show up in CWTFP but not directly in market Gini coefficients. The market Gini measures pre-tax, pre-transfer wage and capital income dispersion; welfare gains from cheaper consumption or better public services are largely invisible to that measure. This measurement mismatch rationalizes why institutions act as a threshold rather than a linear moderator of market-income inequality.

5.4 Synthesis

The mechanism results support Hypothesis 4 and reveal a clear division of labor between the two boundary conditions. Human capital is the amplification mechanism that converts technical suitability into higher-quality employment and deeper GVC integration, directly compressing wage dispersion. Institutional quality is the diffusion mechanism that converts

aggregate productivity gains into consumer welfare, operating through a channel that market Gini statistics do not fully capture. Both mechanisms are consequential for inclusive growth, but they operate through distinct pathways and appear in different dependent variables. Policies targeting inequality through technology need to engage *both* channels simultaneously.

6 Identification and Robustness

The TWFE estimates in Section 4 identify the within-economy, over-time relationship between suitability and inequality after removing fixed country and year effects. A remaining concern is that country-level productivity shocks simultaneously drive suitability (through factor demand) and inequality (through factor prices), biasing the TWFE coefficient toward zero (attenuation) or away from zero (amplification). This section addresses that concern through instrumental variables estimation, lagged dependent variable specifications, and several further robustness checks.

6.1 Instrumental Variables Estimation

Instruments and first stage. We use three sets of instruments for $\ln \text{Suit}_{it}$, following Li et al. [2026]. *Z1 (capital stock lags)*: the third and fourth lags of log capital stock per worker, $\ln k_{it-3}$ and $\ln k_{it-4}$. Physical capital accumulation determined three to four periods earlier shapes the factor-intensity trajectory of technical change today but cannot be caused by contemporaneous distributional shocks. *Z2 (Bartik shift-share)*: a shift-share instrument $\text{Bartik}_{it} = \sum_s \ell_{is,0} \cdot g_{-i,s,t}$, where $\ell_{is,0}$ is economy i 's employment share in sector s in a base year and $g_{-i,s,t}$ is the global (leave-one-out) growth rate of capital intensity in sector s at time t . This instrument isolates variation in suitability driven by global sectoral trends interacted with pre-determined sectoral composition, rather than domestic policy choices. *Z3 (spatial exclusion)*: the log mean suitability of all other economies in the sample, $\ln \overline{\text{Suit}}_{-i,t}$, proxying for technology-frontier diffusion. A single economy's domestic inequality does not plausibly

drive the world technology frontier.

For the moderation specifications, we interact each instrument with the boundary variables ($\ln \text{HC}_{70,i}$, the constant pre-sample measure; and $\mathbf{1}[\text{Double-Low}]_i$). Since these boundary indicators are time-invariant and predetermined, the interaction instruments inherit the excludability of the base instruments.

Table 4 reports 2SLS estimates alongside first-stage diagnostics. Kleibergen–Paap first-stage F -statistics range from 18.4 to 24.7 across specifications, well above the conventional threshold of 10, indicating that the instruments are not weak. Hansen J -statistics fail to reject the overidentification restriction ($p > 0.20$ in all columns), providing no evidence against instrument validity.

2SLS results. Column (1) of Table 4 replicates the baseline without interactions. The 2SLS estimate of β is -0.0228 (se: 0.0091, $p < 0.05$), slightly larger in magnitude than the TWFE estimate of -0.0171 . This upward correction is consistent with classical attenuation bias from measurement error in suitability: TWFE understates the true equalizing effect because the constructed MPK/MPL ratio contains production-function estimation error.

Column (2) introduces the interaction between suitability and the pre-sample human capital proxy. The 2SLS interaction coefficient is -0.0412 (se: 0.0098, $p < 0.01$), close to the TWFE estimate of -0.0381 and retaining 1 percent significance. The stability of the human capital interaction under IV confirms that the amplification finding does not arise from omitted variables that simultaneously raise human capital and depress inequality through channels other than suitability.

Column (3) reports the IV analogue of the double-low threshold test (column 5 of Table 2). The 2SLS coefficient on $\ln \text{Suit} \times \mathbf{1}[\text{Double-Low}]$ is $+0.0134$ (se: 0.0072, $p < 0.10$), again consistent in sign and broadly similar in magnitude to the TWFE counterpart ($+0.0100$). The attenuation-bias-corrected estimate implies that, in doubly disadvantaged economies, the equalizing effect of a one-unit increase in suitability is offset by approximately 1.3

percentage points—about half of the 2SLS baseline effect—further reinforcing the threshold interpretation.

Table 4: Two-Stage Least Squares Estimates: Main Effect and Boundary Conditions

	(1)	(2)	(3)
	Baseline	Human capital	Double-Low
	2SLS	moderator 2SLS	moderator 2SLS
$\ln \text{Suit}$	-0.0228^{**}	-0.0374^{***}	-0.0301^{***}
	(0.0091)	(0.0113)	(0.0108)
$\ln \text{Suit} \times \ln \text{HC}_{70}$		-0.0412^{***}	
		(0.0098)	
$\ln \text{Suit} \times \mathbf{1}[\text{Double-Low}]$			0.0134^*
			(0.0072)
KP first-stage F	24.7	21.3	18.4
Hansen J p -value	0.31	0.24	0.22
Observations	2,438	2,158	2,158
Two-way FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Notes: Instruments for $\ln \text{Suit}_{it}$: third and fourth lags of log capital stock per worker, Bartik shift-share instrument, and log mean suitability of other economies ($\ln \overline{\text{Suit}}_{-i,t}$), following Li et al. [2026]. Columns (2) and (3) additionally use each instrument interacted with the boundary variable as excluded instruments for the interaction term; the boundary variable itself is treated as exogenous. KP = Kleibergen–Paap rk Wald F -statistic for weak identification. Hansen J tests the joint null that all instruments are valid (overidentification test). All regressions include economy and year fixed effects. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. Driscoll–Kraay standard errors in parentheses (bandwidth 3). *, **, ***: 10%, 5%, 1% significance.

6.2 Lagged Suitability Specifications

A further concern is that contemporaneous suitability may partially reflect current-period measurement of factor prices, which are themselves affected by the income distribution. We address this by replacing the contemporaneous $\ln \text{Suit}_{it}$ with lags of one, two, and three years. Lagged suitability is determined before the period inequality is measured, ruling out reverse causation within the lag horizon.

Table 5 reports results. All three lags produce negative, statistically significant coefficients on the main suitability term (columns 1–3). The magnitude of the effect is broadly stable across lag lengths (-0.0163 to -0.0158), consistent with a persistent rather than transitory relationship between suitability and inequality. Columns (4) and (5) confirm that the human capital interaction survives the introduction of a one-year lag: the interaction coefficient remains -0.0374 ($p < 0.01$), essentially unchanged from the contemporaneous estimate of -0.0381 . The stability of the human capital moderator across lag structures rules out the interpretation that the amplification finding is an artifact of high-human-capital economies having systematically lower same-period shocks.

Table 5: Lagged Suitability Specifications

	(1)	(2)	(3)	(4)	(5)
	$L1$	$L2$	$L3$	$L1 + \text{HC}$	$L1 + \text{DL}$
	baseline	baseline	baseline	interaction	interaction
$\ln \text{Suit}_{t-1}$	-0.0163^{**} (0.0075)			-0.0341^{***} (0.0104)	-0.0261^{***} (0.0099)
$\ln \text{Suit}_{t-2}$		-0.0160^{**} (0.0074)			
$\ln \text{Suit}_{t-3}$			-0.0158^{**} (0.0073)		
$\ln \text{Suit}_{t-1} \times \ln \text{HC}_{70}$				-0.0374^{***} (0.0080)	
$\ln \text{Suit}_{t-1} \times \mathbf{1}[\text{Double-Low}]$					0.0094^* (0.0054)
Observations	2,374	2,310	2,246	2,094	2,094
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is $\ln \text{Gini}_{it}$. $L1$, $L2$, $L3$ denote one-, two-, and three-period lags of log suitability, respectively. Column (4) interacts $\ln \text{Suit}_{t-1}$ with demeaned 1970 human capital; column (5) interacts $\ln \text{Suit}_{t-1}$ with the double-low dummy. Controls and fixed effects as in Table 2. Driscoll–Kraay standard errors in parentheses (bandwidth 3). *, **, ***: 10%, 5%, 1% significance.

6.3 Dynamic Human Capital Proxy

The 1970 schooling proxy captures the pre-sample basis of a country’s absorptive capacity but misses the substantial educational expansion that occurred in many economies between 1980 and 2019. As a check, we replace the static 1970 measure with the PWT 10.0 human

capital index [Feenstra et al., 2015]—a time-varying measure constructed from Barro–Lee average schooling and returns to education estimates—restricting the sample to 1990–2019 where the PWT index is consistently available for our economies.

The results are qualitatively and quantitatively similar. The interaction between contemporaneous suitability and the dynamic HC index is -0.0304 (se: 0.0117 , $p < 0.01$). The slightly smaller magnitude relative to the static 1970 estimate is expected: the dynamic index averages over the full within-economy educational distribution and is measured with lower classical error in high-income economies, compressing the cross-economy variation that identifies the interaction. The sign and significance of the human capital amplification effect are preserved, confirming that the finding does not rest on the particular construction of the pre-sample proxy.

6.4 Broader Sample: Unbalanced Panel

Our baseline balanced panel of 64 economies over-represents middle- and high-income countries because low-income economies often lack consistent data for the full 1980–2019 window. This selection may bias estimates toward contexts where human capital and institutions are already adequate, potentially understating how strongly the double-low threshold constrains the equalizing effect of suitability in the poorest economies.

To assess external validity, we construct an unbalanced panel that relaxes the requirement of complete coverage over 40 years, adding 18 additional low- and lower-middle-income economies for the periods in which their data are available. The expanded sample contains 3,814 economy-year observations across 82 economies. Estimating the baseline and moderation specifications on this broader sample yields a suitability coefficient of -0.0154 ($p < 0.05$) in the baseline, and a human capital interaction of -0.0351 ($p < 0.01$). The double-low interaction is $+0.0126$ ($p < 0.05$), somewhat larger than the balanced-panel estimate ($+0.0100$, $p < 0.10$) and now significant at the 5 percent level. This pattern is consistent with the prediction that the threshold effect is particularly consequential in low-income economies,

which are disproportionately represented in the expanded sample. The broader sample thus strengthens the conclusion that institutional quality and human capital jointly constitute a binding constraint on the equalizing power of suitability.

6.5 Alternative Inference Procedures and Winsorization

Alternative standard errors. Table 2 uses Driscoll–Kraay standard errors with bandwidth 3. We also report results under (i) bandwidth 1 and bandwidth 5, (ii) economy-clustered standard errors, and (iii) two-way clustered standard errors (by economy and year). The suitability main effect and the human capital interaction remain significant at conventional levels under all four alternatives. The institutional quality linear interaction remains insignificant, and the double-low interaction retains its positive sign and significance at the 10 percent level or better under all clustering schemes.

Winsorization. To check that results are not driven by outlier economies in the tails of the suitability or Gini distribution, we re-estimate Table 2 with both variables winsorized at the 1st and 99th percentiles. The human capital interaction coefficient moves from -0.0381 to -0.0353 , remaining significant at 1 percent. The double-low interaction moves from $+0.0100$ to $+0.0091$, retaining 10 percent significance. The institutional quality linear interaction remains insignificant throughout. These checks confirm that the main conclusions are not artifacts of influential observations.

Alternative institutional quality measure. The Economic Freedom Index emphasizes market-supporting institutions (property rights, monetary stability, trade openness) and is less informative about redistribution capacity, labor protection, and public service provision. As a check, we replace EFI with the World Bank Control of Corruption Index on the overlapping subsample (1996–2019). The main suitability effect remains negative and significant. The linear institutional interaction again fails to reach significance, and the double-low interaction (now defined using the corruption index) is positive at the 10 percent level. These

results corroborate the threshold rather than linear characterization of institutional quality’s moderating role, and suggest it is not specific to the particular institutional measure used.

6.6 Limitations

Several limitations deserve acknowledgment. *First*, although the IV and lag results are consistent with a causal interpretation, we cannot fully rule out all sources of omitted variable bias. The instruments satisfy the standard exclusion and relevance conditions, but valid instruments are not directly testable. Future work using quasi-experimental variation in technology adoption—such as robot import shocks or digitalization policy reforms—could provide sharper causal identification. *Second*, both boundary conditions are measured with error. The 1970 schooling proxy and the EFI aggregate multiple dimensions into single indices; richer micro-level proxies for worker skill and institutional capacity could sharpen the boundary tests. *Third*, the mechanism regressions establish conditional correlations rather than causal chains. Mediation analysis with valid instruments for the intermediate variables (GVC position, vulnerable employment) would be needed to formally attribute the suitability–inequality channel to GVC upgrading versus employment quality versus welfare diffusion.

7 Conclusion

We examine whether and under what conditions capital-biased technical change suitability—the alignment between the direction of technical change and factor endowments—reduces income inequality across 64 economies from 1980 to 2019. Using two-way fixed effects estimation with cross-sectionally robust standard errors, supported by instrumental variables and lagged specifications, we find that suitability significantly reduces inequality in the baseline specification. But this equalizing effect is conditional on two boundary factors.

Human capital is a quantitatively important amplifier: economies with higher 1970

schooling translate suitability improvements into deeper global value chain integration and higher-quality employment, substantially strengthening the compression of the wage distribution. The amplification survives two-stage least squares estimation and lagged specifications, confirming a robust causal relationship rather than a correlation driven by omitted productivity trends. Institutional quality does not linearly amplify the suitability–inequality channel, but it acts as an indispensable floor condition: when both institutions and human capital are simultaneously weak, the equalizing effect of suitability is significantly attenuated. Mechanism analysis reveals that human capital operates through GVC upgrading and employment quality, while institutional quality operates through consumer welfare channels not fully captured by market Gini statistics.

These findings carry concrete policy implications. *First*, technology-factor alignment—ensuring that innovation trajectories match economy-wide factor endowments—is a necessary but not sufficient condition for inclusive growth. Economies that import or develop capital-intensive technologies without simultaneously investing in labor force education will see productivity gains narrowly appropriated. *Second*, institutional reform and human capital investment are complementary but distinct priorities. Institutional quality matters most as a safeguard against extreme distributional failure; human capital is the active ingredient that converts technical suitability into broad-based wage gains. *Third*, global value chain strategy should be integrated with education policy. The finding that human capital amplifies the value-added production length channel suggests that upgrading industrial capabilities without a skilled workforce is unlikely to generate the distributional gains often attributed to GVC integration.

For rapidly industrializing economies—including China, where the empirical baseline originated—these results support policies that combine capital-intensive technological upgrading with sustained investment in education, vocational training, and social protection. The complementarity between “investing in things” and “investing in people” is not merely an equity objective; it is a prerequisite for the distributional dividends of technical progress

to reach the broad population.

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